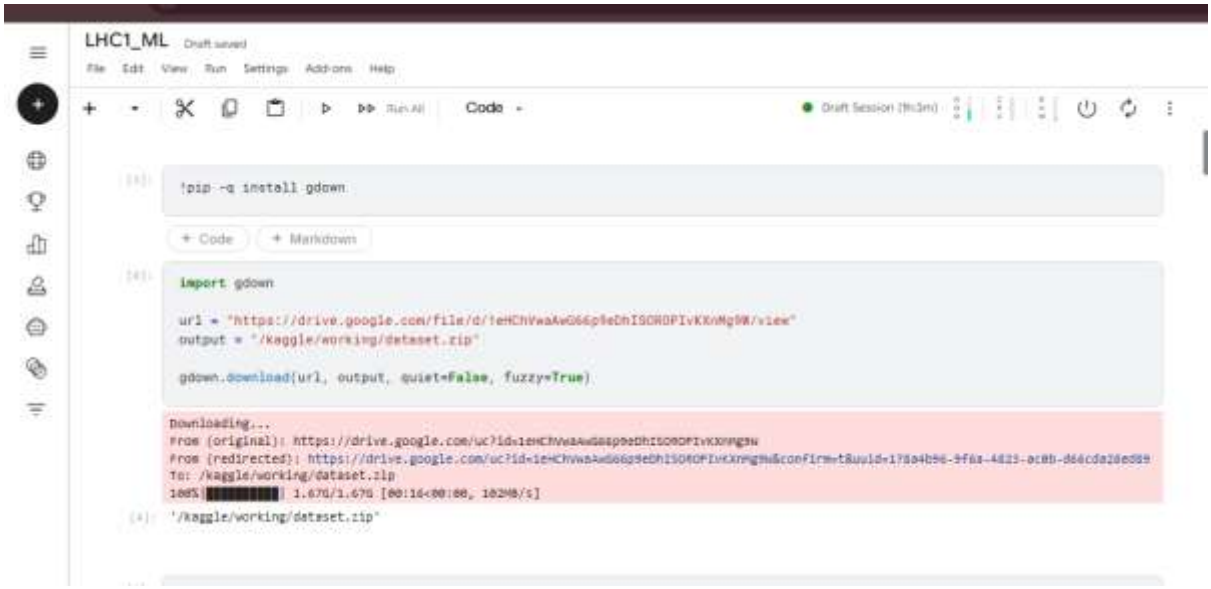
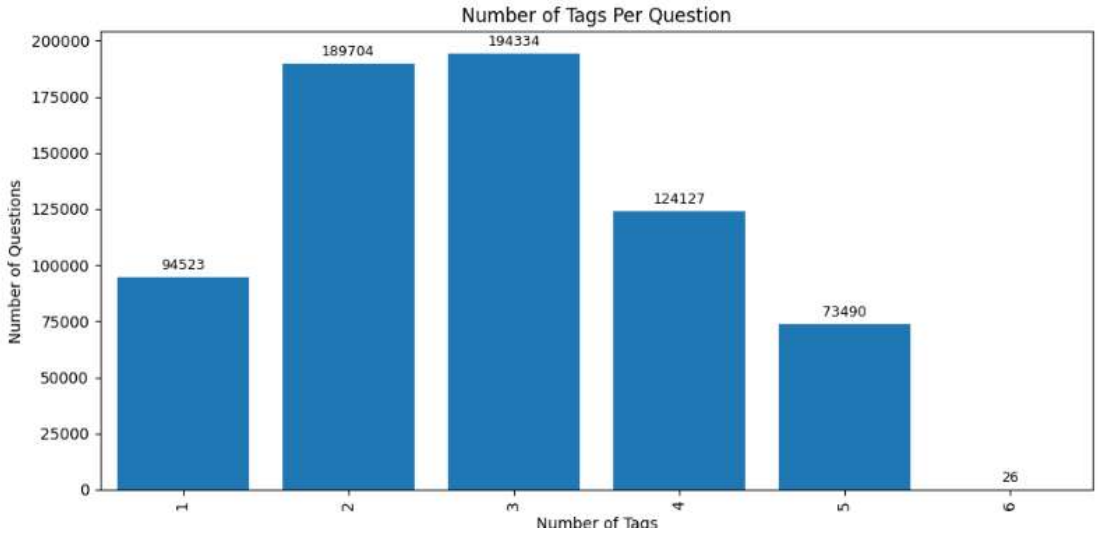


Q1. Number of tags per question

Tags were extracted from each question's tags field and counted. The distribution showed that questions commonly contain two or three tags.



Q2. Total number of unique tags

The dataset contains 25,310 unique tags.

```
# Q2 Determine the Total Number of Unique Tags
import re

all_tags = set()

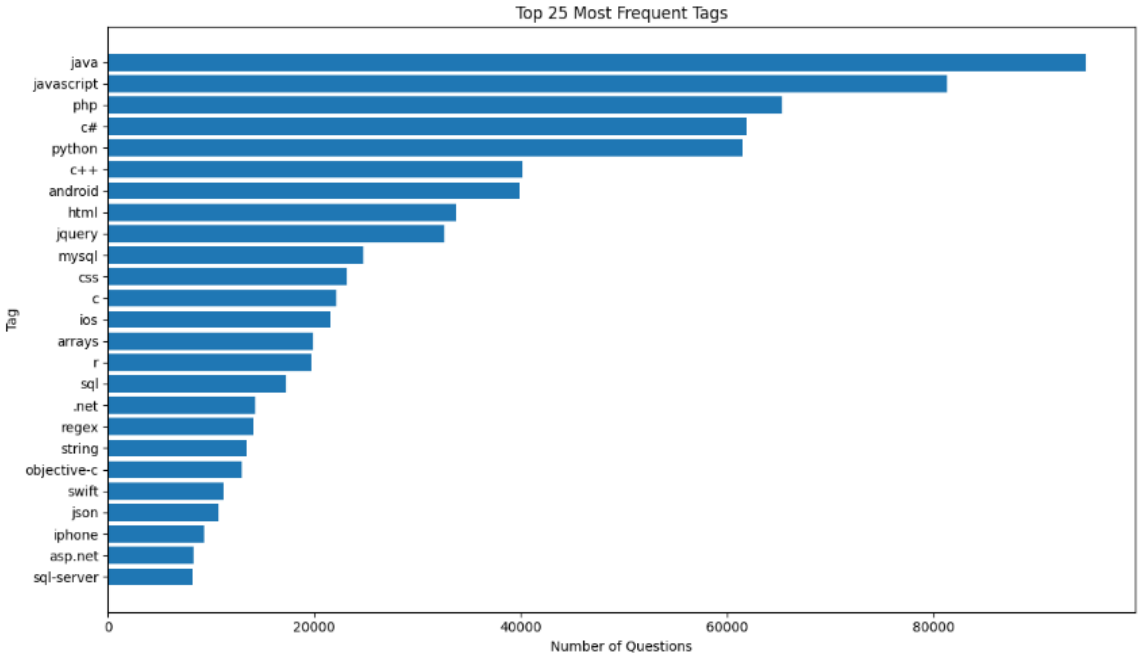
for tag_string in questions["tags"].dropna():
    tags = re.findall(r"<([^>]+)>", tag_string)
    all_tags.update(tags)

print("Total number of unique tags:", len(all_tags))
```

```
Total number of unique tags: 25310
```

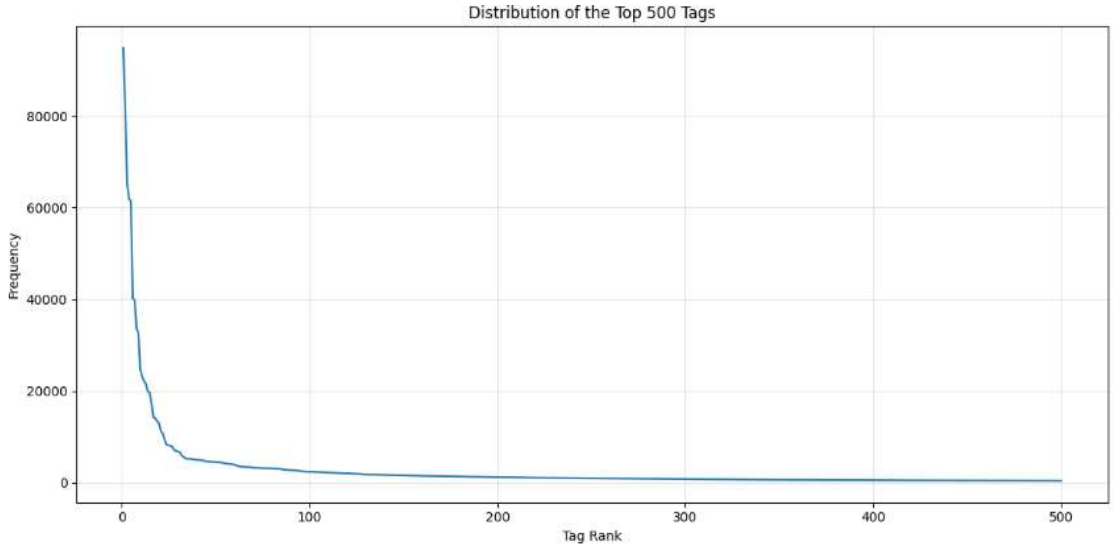
Q3. Top 25 most frequent tags

	Tag	Frequency
0	java	94846
1	javascript	81346
2	php	65333
3	c#	61899
4	python	61505
5	c++	40158
6	android	39910
7	html	33736
8	jquery	32601
9	mysql	24765
10	css	23161
11	c	22166
12	ios	21583
13	arrays	19867
14	r	19744
15	sql	17227
16	.net	14290
17	regex	14107
18	string	13397



Q4. Distribution of the top 500 tags

The frequency plot for the top 500 tags showed a steep decline from the most frequent tags to lower-ranked tags. This indicates an uneven, long-tailed distribution: a relatively small number of tags occur very frequently, while many other tags appear less often.



Q5. Monthly ratio of duplicate questions (2008–2018)

The monthly percentage was calculated as the number of questions whose IDs appeared as duplicate post IDs divided by the total number of questions in that month, multiplied by 100. The observed yearly percentages rose from 5.98% in 2008 to 92.87% in 2018. The monthly line chart also showed a strong upward trend.

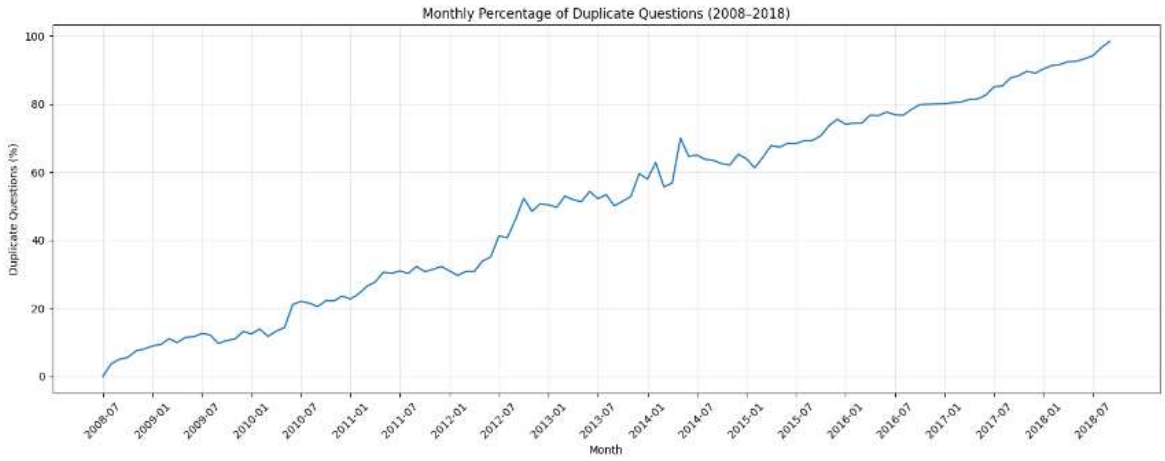
$$\text{Duplicate percentage} = \frac{\text{Duplicate questions in a month}}{\text{Total questions in that month}} \times 100$$

I used

File	Columns used	Purpose
posts_short.csv	id, post_type_id, creation_date	Identifies questions and tells us when each question was created
postLinks.csv	post_id, related_post_id, link_type_id	Identifies links between posts and helps us find duplicate questions

- id: The unique ID of a question in posts_short.csv.
- post_type_id: Identifies the type of post. In our dataset, 1 represents a question.
- creation_date: The date and time when the question was created.
- post_id: The post being linked in postLinks.csv.
- related_post_id: The other post to which it is linked.
- link_type_id: Identifies the type of link. The dataset used 3 for duplicate links.

The dataset contained 682,329 duplicate-link records.However, these are link records, not necessarily 682,329 different questions. One question may have multiple duplicate-link records.



The calculated duplicate percentage showed a general upward trend, increasing from approximately 5.98% in 2008 to 92.87% in 2018. However, further investigation revealed that 231,253 unique IDs marked as duplicates in the duplicate-link dataset were not found in the question dataset, indicating incomplete or mismatched data coverage. Additionally, the percentage reached 98.32% in September 2018, when only 179 questions were recorded. Therefore, these results should not be interpreted as the actual duplicate-question rate on Stack Overflow. The investigation highlights the importance of validating dataset completeness, matching records consistently, and accounting for partial months before drawing conclusions from time-series analysis.

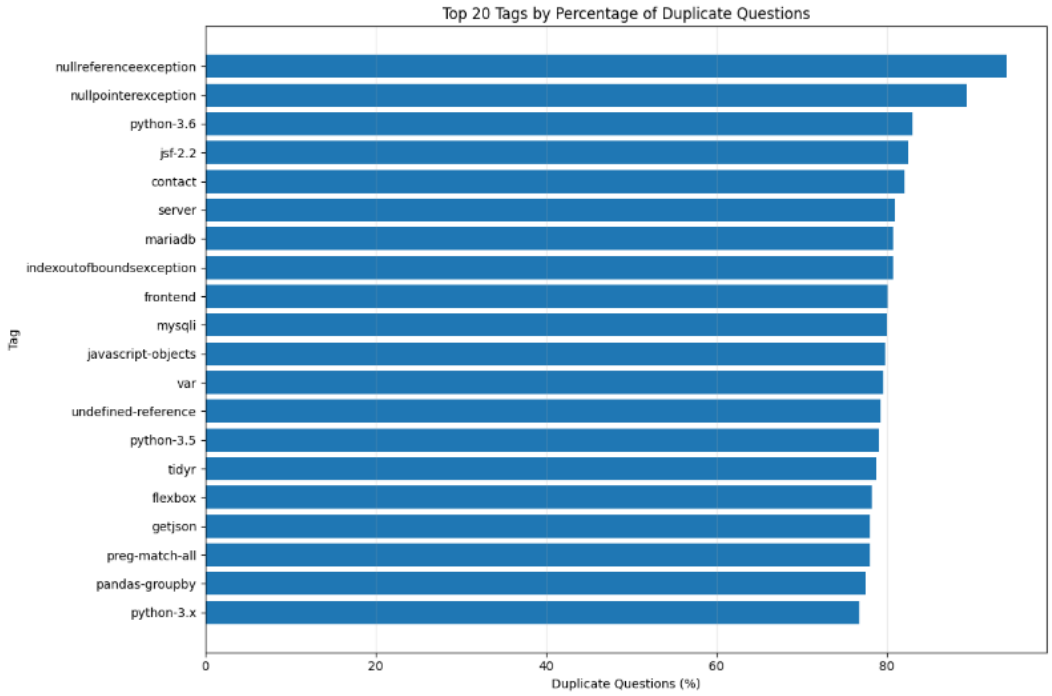
Since a large number of duplicate-question IDs were missing from the question dataset, we should contact the dataset owner to clarify the cause of these missing records and verify the dataset's completeness.

Q6. Duplicate-question percentage by tag

For each tag, the duplicate percentage was calculated as duplicate questions containing that tag divided by all questions containing that tag, multiplied by 100. The top 20 chart was restricted to tags occurring in at least 100 questions to reduce the influence of rare tags. The highest-ranking tags in the chart included nullreferenceexception, nullptrexception, python-3.6, jsf-2.2, and contact. These are associations in the supplied dataset and do not imply that a tag causes duplication.

	tag	total_questions	duplicate_questions	\
3622	nullreferenceexception	505	475	
222	nullpointerexception	2373	2122	
21303	python-3.6	183	152	
16216	jsf-2.2	211	174	
5494	contact	106	87	
6524	server	342	277	
8309	mariadb	156	126	
4280	indexoutofboundsexception	290	234	
1833	frontend	136	109	
433	mysqli	3050	2439	
262	javascript-objects	356	284	
903	var	186	148	
4263	undefined-reference	169	134	
19086	python-3.5	177	140	
14555	tidyr	198	156	
7945	flexbox	1616	1265	
6235	getjson	114	89	
5763	preg-match-all	109	85	
16380	pandas-groupby	111	86	
805	python-3.x	7888	6057	

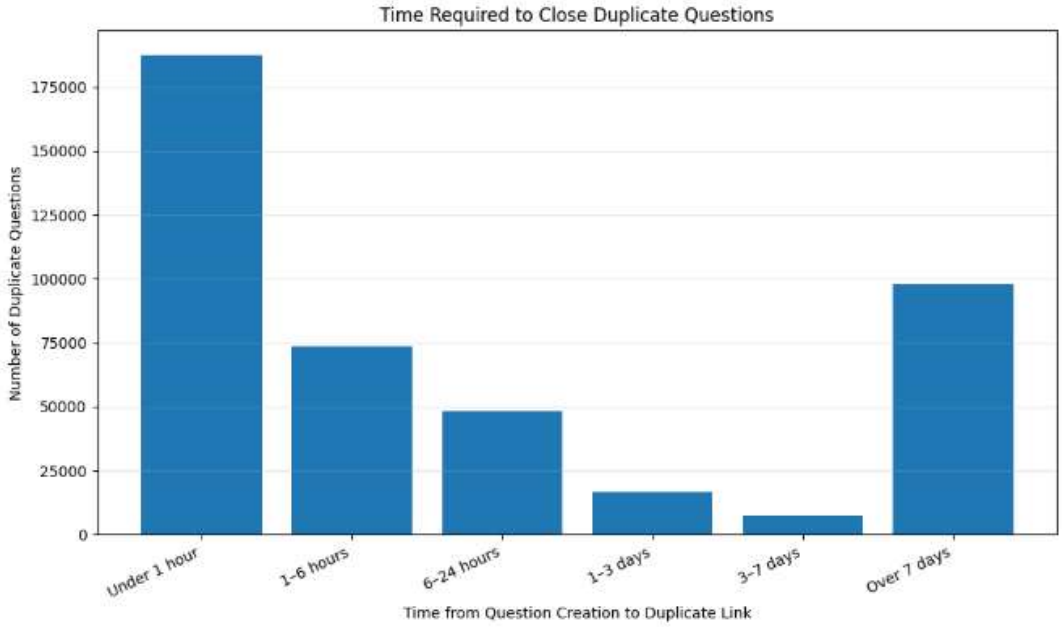
	duplicate_percentage
3622	94.059406
222	89.422672
21303	83.060109
16216	82.464455
5494	82.075472
6524	80.994152
8309	80.769231
4280	80.689655
1833	80.147059
433	79.967213
262	79.775281
903	79.569892
4263	79.289941
19086	79.096045
14555	78.787879
7945	78.279703
6235	78.070175
5763	77.981651
16380	77.477477
805	76.787525



Q7. Time required to close duplicate questions

The chart showed that the largest category was under one hour. The mean was much larger than the median, indicating a right-skewed distribution with some very long intervals. The over-seven-days category covers a broad range of times.

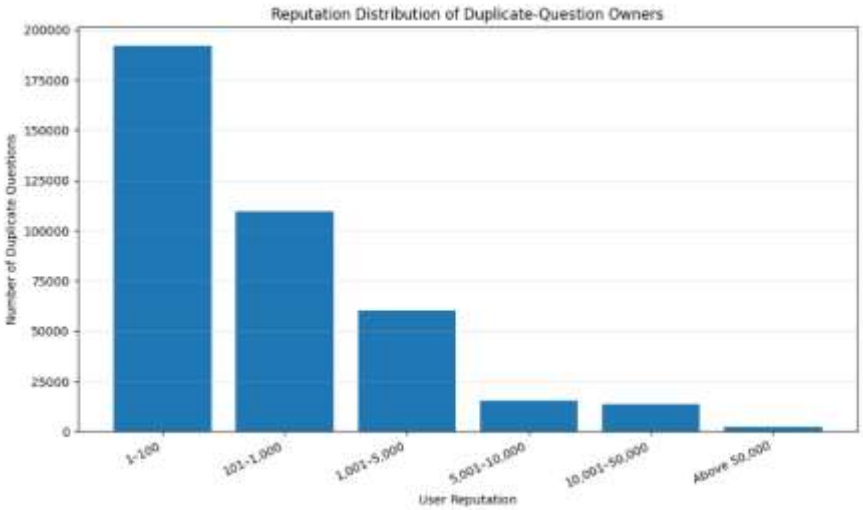
```
Valid duplicate links: 431153
count    431153.000000
mean       2940.065468
std        8458.194047
min           0.001992
25%           0.209015
50%           1.957675
75%          58.200027
max       86959.576839
Name: time_to_close_hours, dtype: float64
```



Q8. Reputation distribution of duplicate-question owners

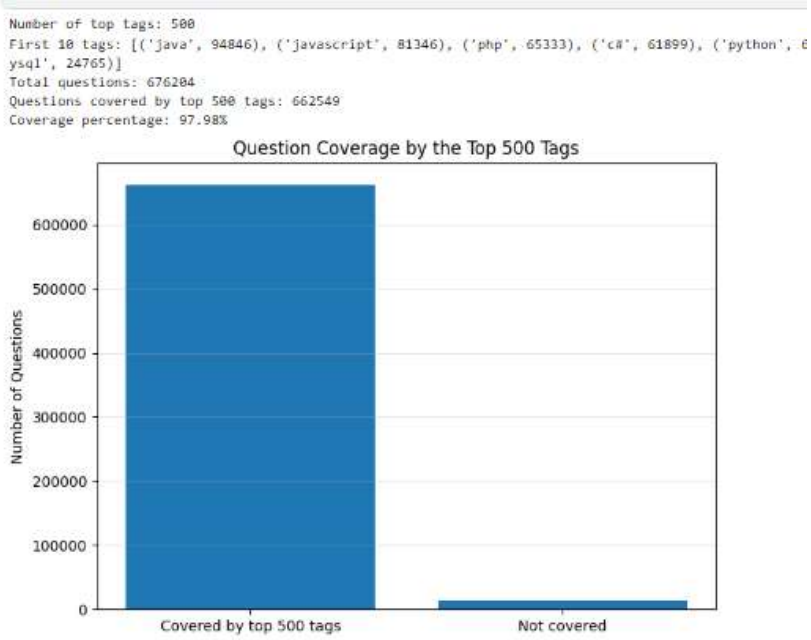
The analysis examined the reputation distribution of users who asked questions marked as duplicates on Stack Overflow. A total of 393,539 duplicate-question records were matched to user reputation data, representing 284,611 unique users. The 1–100 reputation category contained the largest number of question records.

These findings suggest that, in our dataset, duplicate questions were more frequently associated with lower-reputation users. However, the analysis counted question records rather than unique users, so users who asked multiple duplicate questions could be counted more than once.



Q9. Question coverage by the top 500 tags

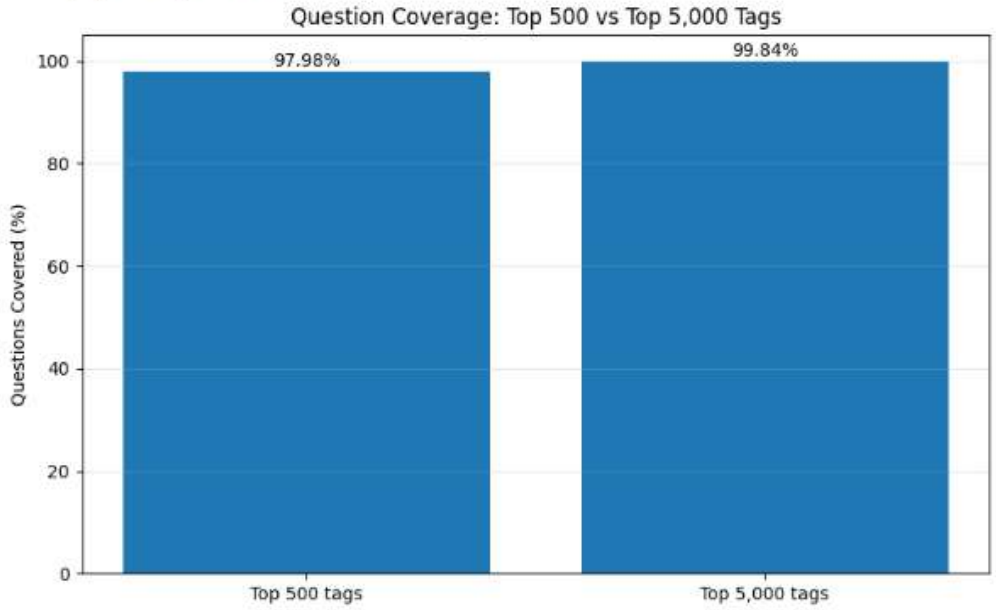
A question was counted as covered if it contained at least one of the 500 most frequent tags. The top 500 tags covered 97.98% of the questions.



Q10. Question coverage by the top 5,000 tags

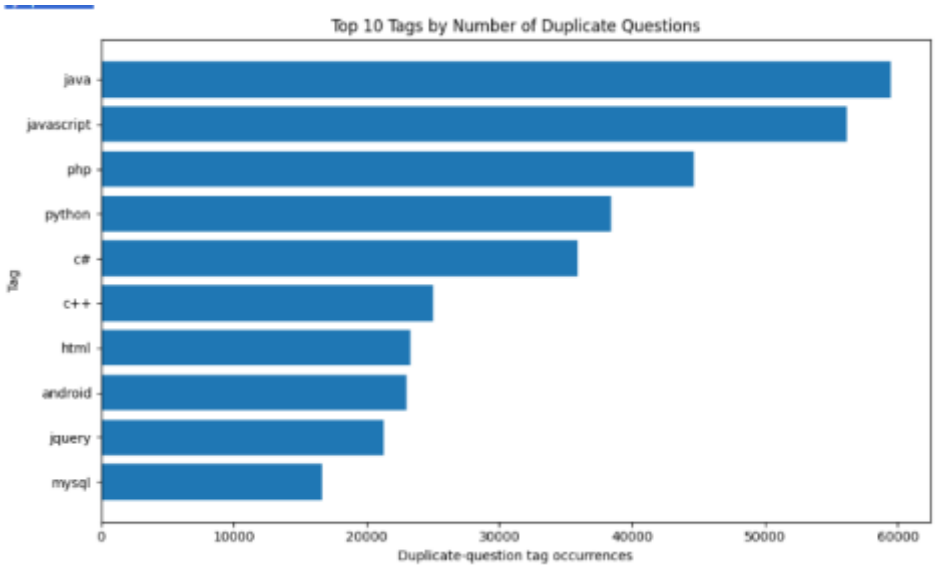
The top 5,000 tags covered 99.84% of questions. Compared with the top 500 tags, coverage increased by 1.86 percentage points and an additional 12,596 questions were covered.

```
Number of top tags: 5000
Total questions: 676204
Questions covered: 675145
Questions not covered: 1059
Coverage percentage: 99.84%
```



Question 11: Top 10 Tags with the Highest Number of Duplicate Questions.

```
java: 59498
javascript: 56178
php: 44619
python: 38427
c#: 35913
c++: 25037
html: 23282
android: 22987
jquery: 21293
mysql: 16641
```

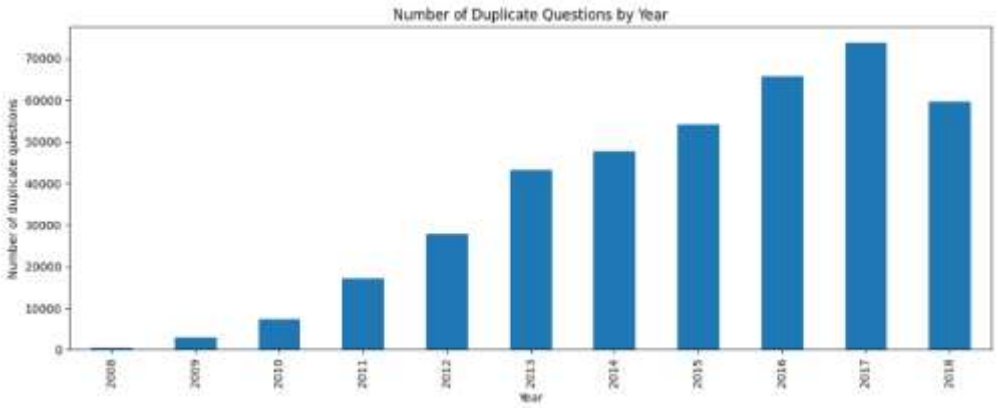


This question to identifies which programming languages, technologies, and topics are most frequently associated with duplicate questions on Stack Overflow. This analysis helps to understand where duplicate-question activity is concentrated and which tags

contribute the greatest number of duplicate-question records.The analysis revealed that java had the highest number of duplicate-question tag occurrences, with 59,498, followed by javascript with 56,178 and php with 44,619. However, these counts do not necessarily mean that these topics have the highest proportion of duplicate questions, because they may also have more questions overall.

Question 12: Yearly Distribution of Duplicate Questions

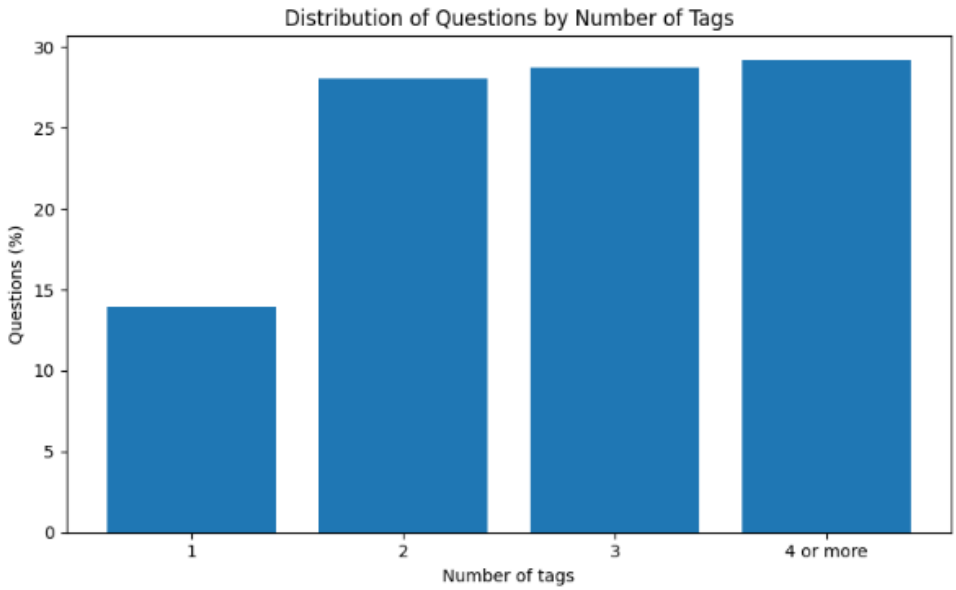
year	duplicate_questions
2008	497
2009	2893
2010	7364
2011	17059
2012	27764
2013	43243
2014	47768
2015	54266
2016	65651
2017	73914
2018	59555



This question investigate how the number of questions marked as duplicates changed over time. A yearly analysis makes it easier to identify long-term trends and understand whether the number of duplicate-question records increased or decreased between 2008 and 2018. The analysis showed a substantial increase in the number of duplicate questions over the study period. The count rose from 497 in 2008 to 2,893 in 2009 and continued increasing, reaching 43,243 in 2013, 65,651 in 2016, and a peak of 73,914 in 2017. In 2018, the count decreased to 59,555. This decline may partly reflect the fact that the dataset ends on September 2, 2018, meaning that the year is incomplete. Overall, the results show an upward trend in the number of recorded duplicate questions. However, as identified in Question 5, incomplete or mismatched dataset coverage means that these counts should not automatically be interpreted as the actual yearly duplicate-question rate across all of Stack Overflow.

Question 13: Distribution of Questions by Number of Tags

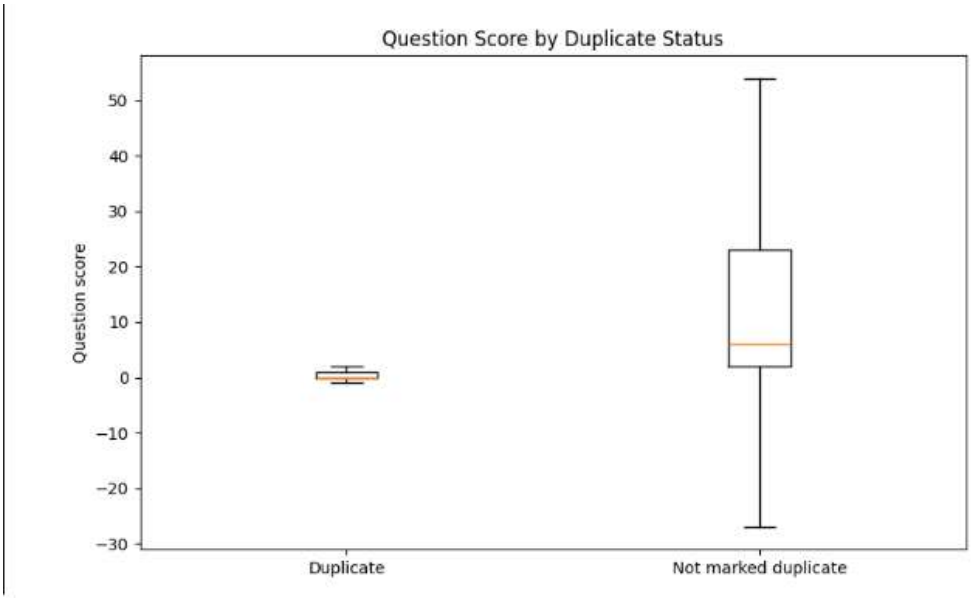
tag_group	question_count	percentage
1	94523	13.98
2	189704	28.05
3	194334	28.74
4 or more	197643	29.23



This question analyses how many tags are typically assigned to a question and how questions are distributed across different tag counts. Although Question 1 calculated the number of tags for individual questions, this question summarises those results into groups and expresses them as percentages, making the overall distribution easier to interpret. The results showed that 94,523 questions (13.98%) had one tag, 189,704 questions (28.05%) had two tags, and 194,334 questions (28.74%) had three tags. A further 197,643 questions (29.23%) had four or more tags. The largest category was therefore questions with four or more tags, closely followed by questions with three tags. These findings indicate that most questions in the dataset contain multiple tags, which can help categorise questions under different topics and make them easier to discover. This analysis complements Question 1 by presenting the tag counts as an overall distribution rather than focusing only on the number of tags attached to each individual question.

Question 14: Question Score by Duplicate Status

status	count	mean	median
Duplicate	399974	1.84	0.0
Not marked duplicate	276230	33.40	6.0

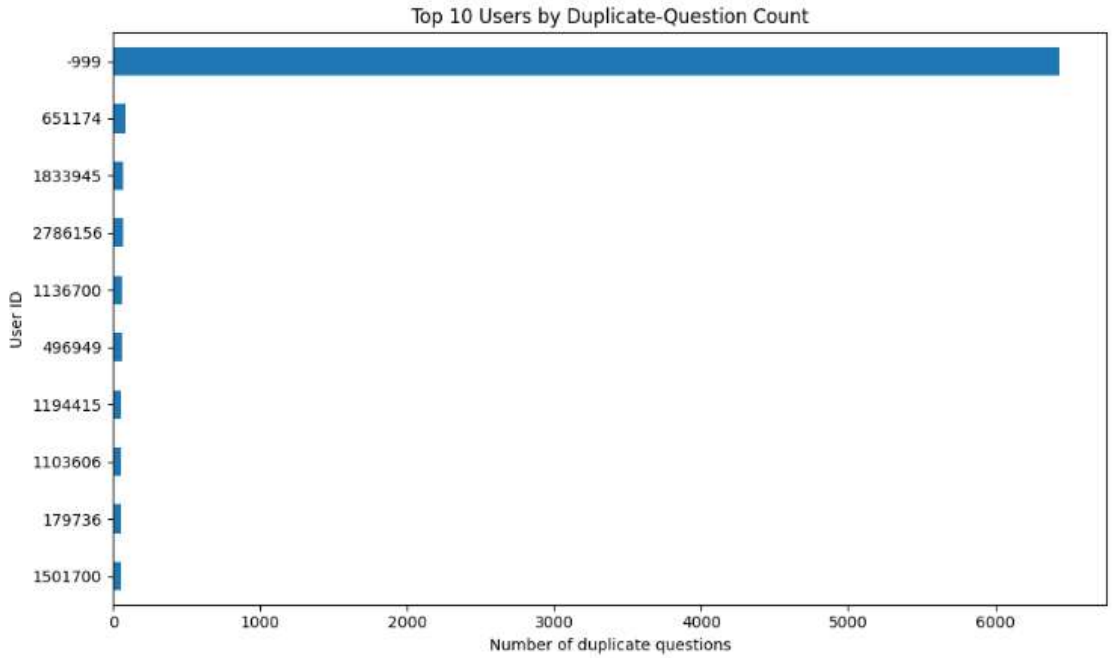


This question investigates whether questions marked as duplicates receive different community voting scores from questions not marked as duplicates. Comparing their scores helps us explore whether duplicate status is associated with differences in community feedback. The analysis compared 399,974 questions marked as duplicates with 276,230 questions not marked as duplicates. The duplicate-question group had a mean score of 1.84 and a median score of 0, whereas the non-duplicate group had a mean score of 33.40 and a median score of 6. The box plot also showed a much wider score distribution for questions not marked as duplicates. These findings indicate that duplicate questions generally had lower scores in the available dataset than questions not marked as duplicates. One possible explanation is that duplicate questions may provide less new information when similar questions already exist. Furthermore, the difference should be interpreted cautiously because the dataset may contain incomplete duplicate-link records.

Question 15: Top 10 Users by Number of Duplicate Questions

Top 10 users by duplicate-question count:

owner_user_id	duplicate_questions
-999	6435
651174	82
1833945	70
2786156	67
1136700	60
496949	58
1194415	55
1103606	54
179736	53
1501700	53



This question identifies the users associated with the greatest number of questions marked as duplicates. This analysis helps explore whether duplicate-question records are distributed evenly among users or concentrated among a smaller number of question authors. The results showed that user ID -999 was associated with 6,435 duplicate-question records, substantially more than any other entry. The next highest counts belonged to user IDs 651174 (82), 1833945 (70), 2786156 (67), and 1136700 (60). The remaining users in the top 10 had between 53 and 58 duplicate-question records. The unusually high count for user ID -999 requires special attention because it may represent an unknown, missing, or placeholder user ID rather than a single identifiable user. Therefore, it should not be interpreted as evidence that one actual user asked 6,435 duplicate questions without further validation. Excluding or separately analysing placeholder IDs would provide a more meaningful understanding of duplicate-question activity among identifiable users. Therefore, we should contact the dataset owner to confirm whether it represents a missing or unknown user ID or a valid user identifier. we should contact the dataset owner to confirm whether it represents a missing or unknown user ID or a valid user identifier.